Matrix Maths of Neural Network

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Matrix Multiplication for H21

Neuron input matrix	Weights matrix
h11 h12	Wh11 Wh21 Wh12 Wh22

h11 × Wh11 Wh21
h12 Wh12 Wh22

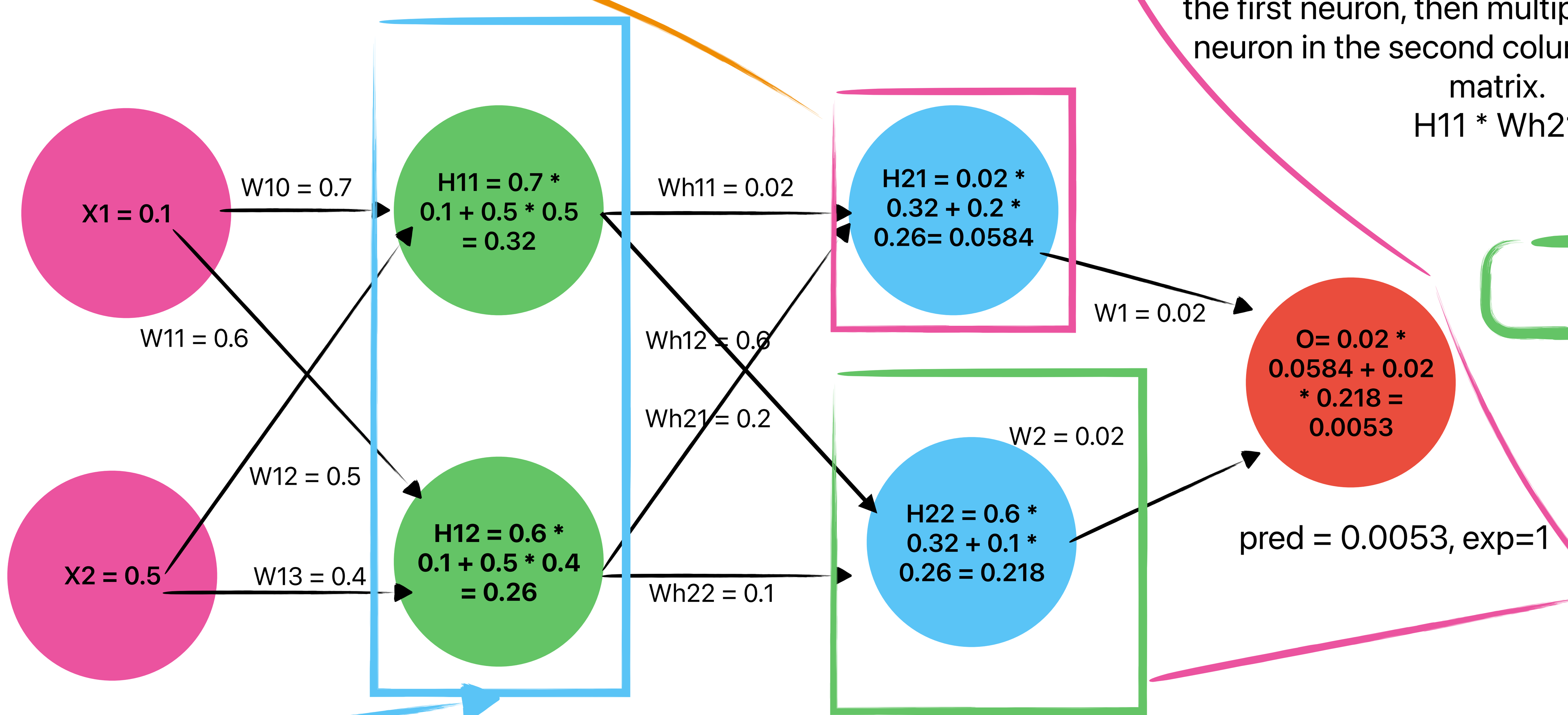
We pick the from the Neuron matrix. We pick the first neuron, then multiply it with the first neuron in the first column in the weight matrix.
H11 * Wh11

h11 Wh11 Wh21
h12 × Wh12 Wh22

Next, we pick the second neuron in the Neuron matrix and multiply it with the second weight in the first column in the weight matrix.
H12 * Wh12

Finally, we add (H11 * Wh11) + (H12 * Wh12)

This will be the input to H21.
H21 = (H11 * Wh11) + (H12 * Wh12)
H21 = 0.02 * 0.32 + 0.2 * 0.26= 0.0584



Matrix Multiplication for H22

Neuron input matrix	Weights matrix
h11 h12	Wh11 Wh21 Wh12 Wh22

H11	×	Wh11	Wh21	=	H21
H12		Wh12	Wh22		H22

h11 × Wh11 Wh21
h12 Wh12 Wh22

We pick the from the Neuron matrix. We pick the first neuron, then multiply it with the first neuron in the second column in the weight matrix.
H11 * Wh21

h11 Wh11 Wh21
h12 × Wh12 Wh22

Next, we pick the second neuron in the Neuron matrix and multiply it with the second weight in the first column in the weight matrix.
H12 * Wh22

Finally, we add (H11 * Wh21) + (H12 * Wh22)

This will be the input to H22.
H22 = (H11 * Wh21) + (H12 * Wh22)
H22 = 0.6 * 0.32 + 0.1 * 0.26 = 0.218

h11 × Wh11 Wh21
h12 Wh12 Wh22
+
h11 Wh11 Wh21
h12 × Wh12 Wh22

Hidden layer 1

Matrix

0.32
0.26

Hidden layer 1

Matrix Weights

One node has one column in the Matrix. So node H11 will be

0.02
0.6

Node H12 will be

0.2
0.1

combined the weight Matrix will be:

0.02 , 0.2
0.6 , 0.1

Hidden layer 1

Matrix

H11
H12

Matrix representation for neurons in a layer is arranged in vertical form. The first neuron comes first and the rest comes underneath.

Hidden layer 1

Matrix for the weights.

To represent the weights of a layer as a Matrix, we will flip the Neuron matrix horizontally.

H11
H12 → H11 H12

Now, each neuron is a column. Each of their weights will come underneath them.

<u>H11 H12</u>
Wh11 Wh21
Wh12 Wh22

Now, this is the weight Matrix for hidden layer 1